

Standards Mitra

React / Next.js Frontend Product Requirements Document

Purpose: Implementation-oriented PRD for a production-ready React/Next.js frontend, based on the provided Standards Mitra frontend PRD.

Target stack: Next.js + React + TypeScript + utility-based CSS + reusable component architecture.

1. Product Vision

Standards Mitra is a chatbot-first web application for Indian Standards, BIS certification schemes, hallmarking, and consumer guidance.

The conversation is the primary UI. The experience should communicate trust through clarity, use minimal chrome, and support accessibility for low-literacy and non-English-first users.

2. Frontend Objectives

Build a responsive frontend that can later connect to backend/API services. Use reusable React components, explicit UI states, consistent tokens, and data-driven repeated elements.

V1 excludes backend/API implementation, full account management, and admin/analytics dashboards.

3. Screen Inventory

01 Desktop Empty State; 02 Desktop Active Conversation; 03 Desktop Conversation + Citation Panel; 04 Mobile Empty State; 05 Mobile Active Conversation; 06 Mobile Citation Bottom Sheet; 07 Voice Input; 08 Photo Scan/OCR; 09 Settings; 10 Loading/Error states.

4. React Component Architecture

```
src/  
■■■ app/ layout.tsx, page.tsx  
■■■ components/layout/ AppShell, Sidebar, SidebarItem, Header  
■■■ components/chat/ Chat, MessageList, UserMessage, AssistantMessage, CitationBadge, TypingIndicator,  
QuickActionChip, ChatInput  
■■■ components/prompts/ EmptyState, PromptCard  
■■■ components/citation/ CitationPanel  
■■■ components/voice/ VoiceOverlay  
■■■ components/photo/ PhotoScanSheet  
■■■ components/settings/ SettingsPanel  
■■■ data/ conversations.ts, prompts.ts, mockMessages.ts  
■■■ lib/ utils.ts
```

Avoid a single monolithic page component. Component boundaries should be understandable in both Figma and React.

5. App Shell

Desktop: sidebar approximately 250–280px, main area, header, centered chat thread, sticky composer. Chat content maximum readable width approximately 700–800px.

Mobile: sidebar becomes drawer, chat becomes full width, composer remains sticky, quick actions scroll horizontally.

6. Sidebar

BIS logo placeholder, Standards Mitra brand, New Chat action, conversation history. Example items: BIS certification requirements; Hallmark verification; IS standard for cement; Consumer complaint guidance; Steel quality standards.

States: default, hover, active, mobile drawer open/closed.

7. Header

Show Standards Mitra, online status, and EN / HI / TE. Desktop height approximately 56–64px. Mobile includes menu/drawer trigger.

8. Empty State

Centered heading: “Ask anything about Indian Standards”. Supporting message explains Indian Standards, BIS certification, hallmarking, and consumer guidance.

Use 3–4 reusable PromptCard components. Example prompts: What is BIS certification? / How do I verify a hallmark? / What does an IS number mean? / How can I check a product standard?

Prompt cards should use a future handler such as onSelectPrompt(prompt).

9. Chat Data Model

```
Conceptual TypeScript model:  
type Message = { id: string; role: "user" | "assistant"; content: string; citations?: Citation[] };  
type Citation = { id: string; standardNumber: string; clause?: string; snippet: string };
```

Use mock data during frontend development. Do not invent API contracts.

10. Chat Components & States

UserMessage renders user content. AssistantMessage renders response content and optional citations. TypingIndicator is visible/hidden. CitationBadge has default/hover/active states.

ChatInput states: empty, typing, focused, disabled, loading. Conceptual handlers: onSend, onVoice, onCamera, onChange.

11. Citation Experience

Clicking a CitationBadge opens CitationPanel. Desktop uses a right-side slide-in panel; mobile uses a bottom sheet.

Panel content: standard number, clause snippet, and “View full standard”. State can include citationPanelOpen and selectedCitation.

12. Voice Experience

Flow: microphone → listening animation → live transcript preview → confirmation/edit → submit.

Use VoiceOverlay with explicit states such as idle, listening, transcript, confirming. Keep it connected to ChatInput.

13. Photo Scan Experience

Flow: camera → capture/upload → preview → scanning → OCR text → confirmation/edit → submit.

OCR text should populate the normal input for user review/editing before sending. Use PhotoScanSheet with explicit states.

14. Settings

Lightweight panel: English / Hindi / Telugu and optional text-size control. Avoid full account/settings management.

15. Design System

Visual direction: neutral white/off-white, dark slate text, restrained navy or muted saffron/blue trust accent, subtle green/blue citation pills, Inter/system sans-serif.

Avoid gradients, neon, excessive shadows, glassmorphism, flashy illustrations, and dashboard-heavy layouts.

Use reusable tokens for colors, typography, spacing and radius. Suggested spacing: 4, 8, 12, 16, 20, 24, 32, 40, 48, 64px. Radius: 8, 12, 16, 24px.

16. Responsive Rules

Desktop: visible sidebar, centered conversation, max width 700–800px, right citation panel, sticky composer.

Mobile: sidebar drawer, full-width chat, sticky composer, horizontal quick-action chips, citation bottom sheet, mobile-friendly voice/photo overlays.

Do not merely scale desktop down; create intentional mobile behavior.

17. Accessibility

Strong contrast, visible focus states, sufficient touch targets, accessible labels for icon buttons, no color-only meaning, readable typography, and clear disabled/loading states. Keep actions obvious and language simple.

18. Frontend State Architecture

```
Potential UI state:  
activeConversation  
messages  
inputValue  
isTyping  
citationPanelOpen  
selectedCitation  
voiceState  
photoScanState  
language  
settingsOpen  
mobileSidebarOpen
```

Use local React state or another lightweight UI-state approach as appropriate. The source PRD does not mandate a particular state library.

19. Figma → React Mapping

```
Sidebar → +  
Header →  
Prompt Card →  
Message Bubble → /  
Citation Badge →  
Citation Panel →  
Quick Chip →  
Input →  
Voice Overlay →  
Photo Sheet →
```

Figma variants should map to React props/state wherever practical.

20. Figma File Organization

01 Foundations; 02 Layout; 03 Components; 04 Desktop Screens; 05 Mobile Screens; 06 Interaction Flows.

Use variables/styles and reusable components from the beginning.

21. Acceptance Criteria

First milestone renders Desktop Empty State with reusable Sidebar, Header, EmptyState, PromptCard and ChatInput.

Active chat supports mock message data, typing state, citations and quick actions. Citation badges open source detail UI. Mobile uses drawer/bottom-sheet patterns. Voice and photo flows have explicit frontend states.

22. Scope Guardrails

Do not add authentication, payments, admin analytics, unrelated account features, or invented backend/API contracts. Use clearly labeled mock data until backend requirements are supplied.

Step-by-Step Process

Step 1 — Foundation

Create design tokens and the 1440×900 desktop shell. Build Sidebar, Header, EmptyState, PromptCard and basic ChatInput. Stop and review before continuing.

Step 2 — Component System

Create reusable Figma components and variants. Name them using their future React component names.

Step 3 — Desktop Active Chat

Add mock user/assistant messages, typing indicator, citations and quick actions. Verify reuse across multiple messages.

Step 4 — Citation Panel

Add desktop right-side source panel and connect the conceptual selectedCitation state.

Step 5 — Mobile

Create mobile empty/active screens, sidebar drawer, sticky composer, horizontal chips and citation bottom sheet.

Step 6 — Voice

Add listening, transcript and confirmation/edit states connected to ChatInput.

Step 7 — Photo/OCR

Add upload, preview, scanning and OCR-review states. Populate ChatInput with extracted text for editing.

Step 8 — Settings & Accessibility

Add language/text-size controls and verify contrast, focus, touch targets and readability.

Step 9 — React/Next.js

Implement the validated design as reusable TypeScript React components. Keep mock data separate from presentation.

Step 10 — QA

Test desktop/mobile breakpoints, loading/disabled states, keyboard/focus behavior, component reuse and visual consistency.

Step 11 — Backend Integration

Only after the frontend is stable, replace mock data with real API integration. Keep API logic outside reusable presentation components.

First Screen to Build: Desktop Empty State

SIDEBAR ≈260px	MAIN AREA
BIS logo Standards Mitra + New Chat Recent conversations	HEADER 56–64px Standards Mitra ● Online EN HI TE

	EMPTY STATE Ask anything about Indian Standards Get clear answers about standards, BIS, hallmarking and consumer guidance. [What is BIS?] [Verify a hallmark] [What is an IS number?] [Product standard?]
	CHAT INPUT [Ask about Indian Standards...] Mic Camera Send

Important: Build only this first screen initially. Once the shell is approved, continue to Step 2 and reuse the same components.